

Effects of cue-synchronized parietal cathodal tDCS on internet gaming disorder: A randomized double-blind sham-controlled trial[☆]

Yixuan Song^{a,b,1}, Yuchen Huang^{a,b,1} , Qihong Zheng^{a,b,1} , Xiaoqin Yang^{a,b},
Yang Guo^{a,b}, Yanan Geng^{a,b}, Huixing Gou^c, Junjie Bu^d, Tianye Jia^e, Guangdong Zhou^f,
Lin Lu^{a,b,g,*}, Jie Shi^{a,b,**}, Yan Sun^{a,b,***}

^a Department of Neurobiology, School of Basic Medical Sciences, National Institute on Drug Dependence, Peking University, Beijing, China

^b Beijing Key Laboratory of Drug Dependence Research, Peking University, Beijing, China

^c Department of Radiology, the First Affiliated Hospital of USTC, Hefei National Research Center for Physical Sciences at the Microscale and School of Life Science, Division of Life Science and Medicine, University of Science and Technology of China, Hefei, Anhui, China

^d School of Biomedical Engineering, Anhui Medical University, Hefei, Anhui, China

^e Institute of Science and Technology for Brain-Inspired Intelligence, Fudan University, Shanghai, China

^f Faculty of Psychology, Tianjin Normal University, Tianjin, China

^g Peking University Sixth Hospital, Peking University Institute of Mental Health, NHC Key Laboratory of Mental Health (Peking University), National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital), Beijing, China

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ABSTRACT

Background and objectives: Internet Gaming Disorder (IGD) is prevalent with limited treatment efficacy. Targeting reducing craving triggered by gaming cues is a critical therapeutic objective. This study aimed to establish optimized electroencephalography (EEG) biomarkers for IGD and develop a novel targeted neuromodulation protocol.

Methods: In the exploratory study, the optimized EEG indicators of IGD diagnose were identified through machine learning models based on event-related potential (ERP) and band power during game cue exposure across two independent datasets (Dataset 1: twenty-five IGD, twenty-two Recreational Game Users, twenty-eight non-gaming Healthy Controls (HC); Dataset 2: twenty-three IGD and twenty-three HC). Subsequently, in the intervention study, a double-blind randomized trial was conducted on forty-six IGD participants to compare active and sham transcranial direct current stimulation (tDCS) targeting the region where the optimized EEG marker was located—central parietal lobe (Pz). Active stimulation (1.5 mA, 20 min, 2 days) was applied during cue exposure (cathode: Pz; anode: right trapezius).

Results: Parieto-occipital P300 (peaked at Pz, IGD > HC) during game reactivity emerged as novel optimized EEG indicators for IGD discrimination (accuracy > 80 %), linked to craving. Then, Pz targeted cathodal tDCS synchronized with game cue exposure could significantly reduce craving ($p < 0.001$), gaming time ($p < 0.001$), and P300 alpha ($p = 0.048$) after intervention and at 1–4 weeks follow-ups, with concomitant improvement of decision-making in the active group. Importantly, these effects generalized to unrepresented gaming cues. Besides, we identified baseline delta power at Pz during gaming cues as a significant predictor for treatment effects.

Conclusion: Our findings establish cue-synchronized tDCS as an effective intervention approach and position the Pz as a novel therapeutic target for IGD neuromodulation.

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^{*} Corresponding author. National Institute on Drug Dependence, Peking University, 38 Xueyuan Road, Haidian District, Beijing, 100191, China.

^{**} Corresponding author. National Institute on Drug Dependence, Peking University, 38 Xueyuan Road, Haidian District, Beijing, 100191, China.

^{***} Corresponding author. National Institute on Drug Dependence, Peking University, 38 Xueyuan Road, Haidian District, Beijing, 100191, China.

E-mail addresses: linlu@bjmu.edu.cn (L. Lu), shijie@bjmu.edu.cn (J. Shi), sunyan@bjmu.edu.cn (Y. Sun).

¹ These authors contributed equally to this work.

1. Introduction

As the global video game market is expected to hit \$522 billion in 2025 and esports becomes an official sport in events like the Asian Games, the growing accessibility and social acceptance of gaming have heightened public health concerns about problematic gaming use [1]. Internet Gaming Disorder (IGD) is defined by persistent and uncontrolled gaming behaviors that result in clinically significant impairment across physical, psychological, and social functioning domains [2]. Diagnostically, while the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) classifies IGD as a condition requiring further research, the International Classification of Diseases 11th Revision (ICD-11) formally codifies gaming disorder as an addictive disorder [3,4]. A hallmark pathological feature of IGD is maladaptive cue-induced craving, driving persistent symptoms and relapse [5]. Affected individuals also demonstrate cognitive deficits, such as impaired impulse control, blunted sensitivity to real-world rewards, and dysfunctional decision-making [6,7]. Neuroscience research has begun to delineate the underlying neurophysiology of IGD, pointing towards dysregulation within brain networks associated with reward processing, inhibitory control, and attention [8]. Specifically, neuroimaging studies have highlighted hyperactivity in regions like the ventral striatum during reward anticipation and cues, alongside hypoactivity in prefrontal control regions during inhibitory tasks [9]. Concurrently, electrophysiological (EEG) studies have identified aberrant event-related potentials (ERPs), such as enhanced P300 amplitudes over parieto-occipital regions in response to gaming cues, suggesting heightened attentional allocation and salience attribution to addiction-related stimuli [10]. This parietal hyperactivity, potentially reflecting impaired top-down regulation from prefrontal areas, is conceptually linked to the core symptoms of craving and loss of control in IGD [11,12]. Epidemiological data reveal a global IGD prevalence of 6.7 % (2024 meta-analysis), with males showing approximately 2.5-fold higher rate than females [13]. Despite its high prevalence and frequent psychiatric comorbidities, IGD frequently goes undiagnosed and untreated. The evidence-based treatment specifically targeting IGD remain underdeveloped [14].

Neuroscience-informed neuromodulation has emerged as a promising approach for IGD treatment. Cue-induced craving, a core IGD feature, can be objectively measured through EEG. For instance, game cues could elicit distinct event-related potentials (ERP) patterns directly linked to prefrontal dysfunction and impaired cognitive control [15]. Also, EEG studies have revealed mixed neural oscillatory patterns in IGD under cue exposure in the parieto-occipital regions [16]. Furthermore, occipital alpha/beta oscillations during cue exposure may correlate with symptom severity in IGD [17]. These EEG signatures enable machine learning (ML) applications, which could effectively distinguish IGD from healthy controls by integrating EEG and clinical features, offering promising biomarkers for craving and severity assessment of IGD [18]. Therefore, advances in EEG monitoring and neuromodulation technologies now enable identification of IGD-specific neuro-electrophysiological signatures, which could not only elucidate the neural mechanisms underlying IGD, but also establish an evidence base for developing precisely targeted neuromodulation interventions.

Transcranial direct current stimulation (tDCS) induces polarization shifts in neuronal membrane potentials via subthreshold electric fields: anodal stimulation enhances the sensitivity of the resting membrane potential through depolarization, while cathodal stimulation reduces neuronal excitability through hyperpolarization [19,20]. While anodal dorsolateral prefrontal cortex (DLPFC)-targeted neuromodulation has proven effective in enhancing executive control and reducing substance consumption and craving in substance use disorders (e.g., alcohol, nicotine, and drugs), its application to IGD remains limited with inconsistent craving reduction outcomes [21–24]. Crucially, the selection of stimulation polarity (anodal vs. cathodal) should be guided by the targeted neural abnormality. Given the observed parietal

hyperactivity in IGD during cue exposure [16,25], which our ML approach aimed to identify and target, we hypothesized that inhibitory (cathodal) stimulation would be most appropriate to normalize this pathological over-activation. This rationale is supported by studies in other addictions, where a decrease in cue reactivity is associated with a reduction in craving [26]. Furthermore, studies have shown that anodal tDCS over the DLPFC can improve IGD symptoms and reduce excessive neural electrical activity in the parietal lobe [11], suggesting a network-level effect but also indirectly supporting the rationale for directly inhibiting the parietal node. Therefore, it is essential to identify specific neural electrical alterations in the cerebral cortex of individuals with IGD and apply corresponding excitatory or inhibitory modulation. This approach is also important for exploring the related brain functional networks in IGD.

Therefore, this study pursued two primary objectives: firstly, to identify EEG biomarkers for IGD diagnosis and craving using ML, and secondly, to evaluate intervention effects of tDCS targeting these identified neural signatures. We acquired resting-state and cue-elicited EEG data from individuals with IGD, recreational game users (RGU, which represents a “risk group” between addicted and healthy states), and non-gaming healthy controls (HC), alongside assessments of addiction severity and game craving. The ML model was trained and validated across two independent datasets to detect the optimized cortical EEG features linked to IGD diagnosis and craving. Subsequently, we conducted a double-blind randomized controlled trial (DBRCT) for subjects with diagnosed IGD to assess whether tDCS modulation of the neural activity of proposed could reduce craving and ameliorate cognitive dysfunction. Together, these findings establish a framework for biomarker-guided diagnosis and precision neuromodulation in IGD.

2. Materials and methods

2.1. Design and procedures

There were two parts in this study. In the exploratory study (part 1), we identified optimized EEG indicators of IGD diagnosis and craving through an EEG-based ML model during game cue exposure across two independent datasets. In the intervention study (part 2), the therapeutic potential of tDCS on IGD targeting the optimized EEG indicator (derived from the ML model) was assessed by conducting a DBRCT.

Procedures of the exploratory study: To explore the optimized location and indicator for neuromodulation of IGD, the baseline data for three groups (IGD, RGU and HC) were collected, including demographic/gaming questionnaires, closed-eye resting-state EEG (5 min), and task-state EEG (only GCRT). EEG was acquired using a 32-channel mbt Smarting system (10–20 system, impedance <5k Ω , 500Hz sampling) with participants being instructed to minimized movement/artifacts (Fig. 1D).

Procedures of the intervention study: The intervention trial spanned four consecutive in-lab days plus follow-ups assessments conducted at precisely post_week 1 (Day 7 $\pm$ 1), post_week 2 (Day 14 $\pm$ 2), post_week 3 (Day 21 $\pm$ 2), and post_week 4 (Day 28 $\pm$ 2) after the completion of the second (and final) tDCS session. (Fig. 1E). Day 1 collected baseline data: demographic/gaming questionnaires, followed by closed-eye resting-state EEG (5 min) and task-state EEG (including GCRT, SST, MID, and IGT). Days 2–3 delivered active/sham tDCS synchronized with game cues for 2 consecutive days; pre/post stimulation assessed game craving, mood, side effects, and sensation of current intensity. Day 4 repeated post-intervention questionnaires, resting EEG, and task EEG. Follow-ups occurred at post_week 1 and post_week 4 (in-lab: craving/mood questionnaires, resting/task EEG) and post_week 2 and post_week 3 (online: weekly gaming characteristics/craving/mood assessments) (Fig. 1D and E).

This study was reviewed and approved by the Peking University Institutional Review Board (IRB00001052-24146) and was performed following the relevant guidelines and regulations in two cities in China

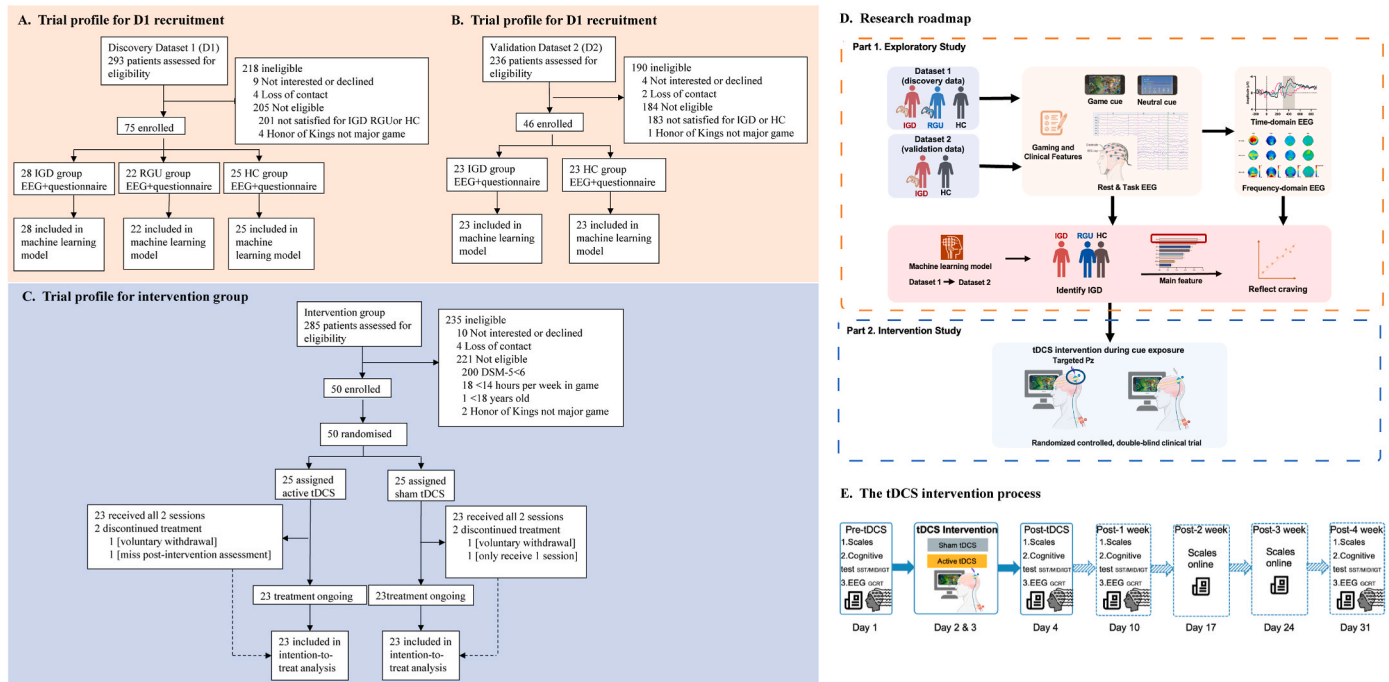


Fig. 1. Trail profile and Experimental procedure. The exploratory study in D1 (A) and D2 (B). (C) The intervention study. (D) Research roadmap. (E) The tDCS intervention process. IGD: Internet gaming disorder; RGU: recreational game user; HC: healthy control; tDCS: transcranial direct current stimulation.

(Beijing and Tianjin). The DBRCT was also registered with [ClinicalTrials.gov](https://clinicaltrials.gov), NCT06759051.

2.2. Participants

Subjects were recruited via online advertisements, including individuals with Internet gaming disorder (IGD), recreational game users (RGU), and healthy controls (HC). All subjects were required to be aged between eighteen and twenty-five and signed an informed consent form and voluntarily participated in this trial. A detailed description of the inclusion and exclusion criteria, please see the Supplemental Methods.

The exploratory study consisted of two datasets: Discovery Dataset 1 (D1), all subjects were recruited from Tianjin between April 11, 2024 and June 20, 2024, including twenty-five in the IGD group, twenty-two in the RGU group, and twenty-eight in the HC group; Validation Dataset 2 (D2), the IGD group was all from Beijing, with a total of twenty-three subjects, and the HC group was all from Tianjin, with a total of twenty-three subjects. The subjects in intervention study were all IGD, and a total of forty-six subjects were recruited from universities in Tianjin ($n = 23$) and Beijing ($n = 23$, the same participants in IGD group of D2), which began on January 7, 2025 and continued through February 15, 2025 (Fig. 1).

2.3. Demographic information and questionnaires

Demographic information was collected by self-reported included the subject's age, education, health status, ethnicity, medication use, and acceptance of weak current stimuli. Gender was self-reported and then verified by researchers for concordance with sex assigned at birth. Addiction severity degree was assessed by DSM-5 diagnostic criteria for IGD and the 9-item Internet gaming disorder scale-short-form (IGDS9-SF) [27]. Game craving was assessed by the Questionnaire of Gaming Urges-Brief (QGU-B) and the visual analogue scale (VAS) with a straight line with "0" at one end indicating "no craving at all" and "9" at the other end indicating "craving very much" [28].

The depression symptoms and anxiety symptoms were assessed by the Patient Health Questionnaire-9 (PHQ-9) and the Generalized

Anxiety Disorder-7 (GAD-7), respectively. The subject's emotional state (positive and negative mood) was assessed using the Positive and Negative Affect Schedule (PANAS) [29].

2.4. Cognition paradigm testing

The following tasks were under EEG (for the more details, please see the Supplemental Methods).

- 1) The Game Cue Reactivity Task (GCRT) primarily tests the subjects' craving level for game-related cues; The exploration part includes thirty positive game cue pictures, thirty negative game cue pictures, and thirty neutral pictures; the intervention part includes fifty positive game cue pictures and thirty neutral pictures. During the tDCS intervention, only positive game cue pictures are presented, with a total of thirty shown (presented cues). Notably, there were twenty positive game pictures which were unrepresented in GCRT were used to assess the generation effects of the intervention on craving.
- 2) The Stop Signal Task (SST) is a classical experimental paradigm that assesses attentional and inhibitory control by measuring an individual's ability to inhibit an initiated action during a task [30].
- 3) The Monetary Incentive Delay (MID) task is a common paradigm for examining reward and punishment processing during anticipation and response [31].
- 4) The Iowa Gambling Task (IGT) is a test where people make choices from card decks with hidden risks and rewards. It measures how well someone makes real-life decisions involving uncertainty and consequences [32].

2.5. tDCS intervention

2.5.1. Sample size calculation in intervention part

Power analysis targeting the group $\times$ time interaction effect was performed using G*Power 3.1.9.7. For a 2×7 repeated measures ANOVA with a medium effect size ($f = 0.25$), thirty-two participants (sixteen per group) were required to achieve 95 % power at $\alpha = 0.05$. This assumed a correlation of $\rho = 0.5$ among repeated measures and

nonsphericity correction $\epsilon = 0.75$ based on Mauchly's test results from pilot data. To accommodate 20 % attrition, at least twenty participants per group were enrolled.

2.5.2. Randomization and blinding

Participants were randomly assigned (1:1) to active or sham tDCS groups using a computer-generated randomization sequence. The allocation was performed by a single, unblinded researcher who prepared the stimulation files on the tDCS device, labeling them with anonymous participant identifiers only. To ensure blinding, the treating investigators and participants remained unaware of the group assignment. The operators administered stimulation by selecting the pre-loaded file corresponding to the participant's ID without knowledge of the stimulation parameters. Participants were not informed of their assignment, and the device was positioned behind them to obstruct any view of the screen or display.

2.5.3. tDCS intervention procedures

tDCS was delivered using the DC-STIMULATOR MC (Neuroconn, Germany). The cathode (5×5 cm) was placed at Pz (parietal cortex), and a matching anode on the right trapezius muscle. This extracephalic montage is an established approach for maximizing focality under the cathode and minimizing unintended modulation of other brain regions [33,34]. Active stimulation consisted of a 20-min protocol: 10-s ramp-up to 1.5 mA, 19 min 40 s at 1.5 mA, and 10-s ramp-down. Sham stimulation replicated the initial/final 20-s ramp-up/down sensations (to 1.5 mA and back to 0 mA), but maintained 0 mA current for the intervening 19 min 20 s, mimicking skin sensation without cortical modulation.

To validate the focality and efficacy of this montage, we performed finite-element modelling (FEM) using SimNIBS 4.0 (64-layer head model, 1 mm resolution). The simulation confirmed a focal peak electric field (0.26 V/m) on the cortical surface beneath the cathode at Pz—exceeding the empirical threshold for neuronal modulation [35]—with negligible off-target stimulation (<0.1 V/m in frontal or temporal regions).

2.5.4. Blinding efficacy and safety assessment

To evaluate blinding efficacy and intervention safety, participants completed a questionnaire after every intervention session, reporting on subjective current intensity perceptions and any discomfort experienced during stimulation. Perceived intensity was measured using a scale adapted from Fertonani et al. [36], ranging from 0 (“None—I did not feel the described sensation”) to 5 (“Strong—I strongly felt the described sensation”), with intermediate levels of very mild (1), mild (2), moderate (3), and considerable (4). Blinding efficacy was assessed by comparing self-reported current intensity score between the active and sham groups using independent samples t-tests; the absence of a significant difference ($p > 0.05$) was interpreted as successful blinding. This objective approach—focusing on sensory experience rather than subjective guesses—was deliberately chosen to avoid cognitive bias and is well-validated in tDCS literature [37–39].

2.5.5. Outcomes measures

The primary outcome was VAS-assessed game craving. Secondary outcomes included QGU-B assessed game craving, daily gaming time, P300 component changes at Pz and cognitive task performance (SST, MID, IGT metrics). Adverse events were assessed during/post tDCS, including fever, burns, development of drowsiness, and so on. Data collected in both Beijing and Tianjin were centrally assessed.

2.6. Statistical analysis

2.6.1. Behavioral data analysis

Statistical analyses used GraphPad Prism 9 (significance: $p < 0.05$, two-sided). Demographic variables (gender, age, playtime, QGU-B, VAS, DSM-5, IGDS9-SF, PHQ-9, GAD-7) were compared with ANOVA (three-

group comparisons in D1) or t-tests (D2, intervention groups) for continuous data, and chi-square for categorical data. Age and gender were included in the analysis as covariates to control for potential confounding effects. Exploratory analyses employed two-way ANOVA for craving: Group (IGD/RGU/HC) $\times$ Cue-Type (Positive/Neutral/Negative Game Cues) with Šídák post-hoc. Pre-intervention analyses included: one-way ANOVA for cue-exposure craving (Neutral/Active-Presented/Active-Not-Presented cues) with Šídák post-hoc; independent t-tests for baseline SST (Go accuracy, Go RT, Stop accuracy, and SSRT) and MID metrics (RT/d' across conditions) between active/sham groups. Longitudinal intervention effects used two-way RM-ANOVA with Group (active/sham) $\times$ Time (pre/post/1wk/4wk) for: mood (PANAS, PHQ-9, GAD-7), craving (QGU-B, VAS), playtime, SST/MID metrics, and IGT net scores across stages (6×10 trials), with Šídák post-hoc identifying significant time points and group differences.

2.6.2. EEG analysis

EEG data were preprocessed in EEGLab/FieldTrip (MATLAB 2021b), preprocessing entailed: (i) positioning the electrodes in accordance with the international 10–20 system to guarantee spatial fidelity; (ii) band-pass filtering the continuous EEG between 0.1 and 30 Hz to retain the dominant neurophysiological frequency bands; (iii) epoching the task-based EEG data into -0.5 to $+1.5$ s segments time-locked to stimulus onset for the task condition, and 2s segments for the resting-state recordings; resting-state EEG was epoched into 2-s segments; (iv) visual inspection and manual rejection of artifact-contaminated epochs, with a pre-specified exclusion criterion of ≤ 80 % of the total number of segments per participant—subjects exceeding this threshold were discarded; (v) spherical-spline interpolation of malfunctioning channels exhibiting sustained high-amplitude deviations; (vi) re-referencing the data to the averaged mastoids (TP9/TP10) to attenuate movement- or impedance-related drifts; and (vii) performing extended-infomax independent-component analysis (ICA) to identify and remove ocular (EOG) and muscular (EMG) artifacts, with the corresponding independent components subtracted from the concatenated dataset.

The resulting power spectral density (PSD) was logarithmically transformed to yield decibel power spectral density (dBPSD). Time-frequency analysis employed the multitaper convolution method combined with a Hamming window, with results output as PSD. Frequency analysis (delta:1–4Hz, theta:4–8Hz, alpha:8–13Hz, beta:13–30Hz) and ERP analysis (P300: 300–500 ms) were performed. Group comparisons used repeated-measures ANOVA: for resting-state/P300, between-group factors were subgroups (D1: IGD/RGU/HC; D2: IGD/HC) with electrode sites as the within-group factor; for intervention effects on cue-exposure P300 at Pz, the between-group factor was treatment (active/sham) with time (pre/post/1-week/4-week) as the within-group factor (Šídák post-hoc tests). Spearman correlations examined associations between: 1) Pz P300 amplitude changes and craving changes post-intervention, and 2) baseline Pz P300-band power and post-intervention craving reduction.

2.6.3. Machine learning

For machine learning algorithms (part 1), Support Vector Machines (SVM) with a linear kernel (MATLAB 2021b) were used to classify IGD vs. HC/RGU groups by finding an optimal hyperplane. The modeling procedure was conducted in three sequential phases to ensure robust feature selection and generalizability. First, within Dataset 1 (D1), we constructed separate machine learning models for each comparison—IGD vs. RGU and IGD vs. HC—using features derived from both resting-state and task-based EEG data. The performance of all models was rigorously evaluated using leave-one-out cross-validation (LOOCV). Subsequently, a cross-dataset validation was performed by training a model on the task-based EEG data from D1 for IGD vs. HC classification and testing its performance on the independent Dataset 2 (D2). Finally, to further confirm the stability of the task-based features, an additional IGD vs. HC model was developed and evaluated internally within D2 using LOOCV.

The model incorporated features from resting-state EEG (energy of four frequency bands across thirty-one electrodes) and the P300 time window (power spectral density at eight electrodes). Hyperparameters, including the regularization parameter C , were optimized. Model performance was evaluated using accuracy, sensitivity (IGD identification), specificity (control identification), and AUC. Feature importance was analyzed by extracting SVM weight vectors across cross-validation rounds to identify high-contribution electrodes, followed by Spearman correlation analysis with clinical craving metrics.

3. Results

3.1. Participant characteristics of the exploratory study

The exploratory study comprised two datasets: Discovery Dataset 1 (D1), which included 25 (48 % female) participants in the IGD group, 22 (50 % female) in the RGU group, and 28 (57.14 % female) in the HC group; and Validation Dataset 2 (D2), which included 23 (47.83 % female) IGD individuals and 23 HC (52.17 % female) individuals (Fig. 1A and B). Sex distribution was comparable. Age differed significantly among IGD (21.60 ± 2.00), RGU (20.59 ± 0.85) and HC (20.43 ± 0.50) groups in D1 ($F(2,72) = 6.24, p = 0.003$) and between IGD (22.04 ± 0.50) and HC (20.43 ± 0.10) groups in D2 ($t(44) = 3.14, p = 0.003$). All gaming measures, including daily gaming duration, QGU-B craving scores, IGDS9-SF scores, and DSM-5 criteria, showed significant group differences (all $p < 0.001$ in either D1 or D2). Besides, PHQ-9 and GAD-7 scores also differed significantly among groups in two datasets (all $p < 0.017$) (Table 1). Sex and age were included as a covariant in the following analysis.

3.2. EEG characterization of IGD

In D1, resting-state EEG analysis revealed significant group differences in the whole-brain delta ($F(2, 2139) = 22.42, p < 0.001$), theta ($F(2, 2139) = 30.15, p < 0.001$) and alpha ($F(2, 2139) = 20.78, p < 0.001$) and band power spectral density (PSD) (Supplementary Fig. 1).

During gaming cue reactivity tasks, IGD participants exhibited significantly higher craving scores compared to participants from both HC ($p < 0.001$) and RGU groups ($p = 0.003$) under positive gaming cues. IGD patients also had higher craving scores for positive cues than for neutral cues ($p < 0.001$) or negative cues ($p < 0.001$) (Fig. 2A). These results hence indicated that positive gaming cues specifically amplify craving in IGD.

Parieto-occipital P300 component under positive cues differed significantly between IGD, RGU and HC ($F(2, 576) = 31.81, p < 0.001$). Post-hoc analyses confirmed higher P300 amplitudes in IGD versus HC ($p < 0.001$) (Fig. 2B). Analysis of P300 time-window PSD revealed group differences in theta ($F(2, 576) = 6.27, p = 0.002$) and alpha ($F(2, 576) = 3.62, p = 0.028$) bands (Fig. 2C).

Consistent with findings in D1, parieto-occipital P300 mean amplitudes were significantly higher in IGD versus HC ($F(1, 352) = 174.60, p < 0.001$) in D2 (Supplementary Fig. 2A). PSD analysis identified group differences in delta ($F(1, 352) = 22.76, p < 0.001$) and theta ($F(1, 352) = 37.91, p < 0.001$) bands. (Supplementary Fig. 2B).

3.3. Parieto-occipital P300 response to positive game cues optimally identified IGD and correlated with craving

SVM models using resting-state whole-brain spectral features achieved moderate-to-low accuracy (66.0 %) in distinguishing IGD from HC in D1 (sensitivity: 70.8 %, specificity: 61.5 %, AUC: 0.718). In contrast, models incorporating time-window PSD features of parieto-occipital P300 under positive cues achieved superior accuracy (accuracy: 81.1 %, sensitivity: 80.0 %, specificity: 82.1 %, AUC: 0.810) (Fig. 3A). Similarly, classification of IGD versus RGU was more accurate using cue-induced P300 features than resting EEG features (AUC: 0.681 vs. 0.609) (Fig. 3B).

Table 1
Demographic information and game use characteristics.

	Dataset 1 of the exploratory study					Dataset 2 of the exploratory study					Intervention groups					
	HC (n = 28)		RGU (n = 22)		IGD (n = 25)	p	Statistics		HC (n = 23)	IGD (n = 23)	p	Statistics	Active (n = 23)	Sham (n = 23)	p	Statistics
Age (years)	20.43 (0.50)	20.59 (0.85)	21.60 (2.00)	21.60 (2.00)	0.003	F = 6.24	20.43(0.10)	0.003	22.04(0.50)	0.003	t = 3.14	20.78(0.41)	21.09(0.51)	0.643	t = 0.46	
Female/Male (Female%)	12/16 (57.14 %)	11/11 (50.00 %)	13/12 (48.00 %)	13/12 (48.00 %)	0.78	$\chi^2 = 0.49$	11/12(52.17 %)	0.77	12/11(47.83 %)	0.77	$\chi^2 = 0.09$	9/14(39.13 %)	10/13(43.48 %)	0.768	$\chi^2 = 0.09$	
Daily gaming time (hours)	0.88 (0.22)	3.59 (1.22)	4.32(1.65)	4.32(1.65)	<0.001	F = 64.33	0.89(0.04)	<0.001	2.90(0.21)	<0.001	t = 9.49	3.02(0.23)	3.18(0.22)	0.622	t = 0.32	
Year of regularly playing games (years)	15.59(0.56)	13.95(0.79)	14.94(0.99)	14.94(0.99)	0.266	F = 1.35	15.69(0.79)	0.878	15.52(0.76)	0.878	t = 0.16	15.39(0.72)	15.09(0.83)	0.783	t = 0.28	
Primary motivation	Achievement	4	2	3	0.920	$\chi^2 = 1.99$	4	0.171	6	0.171	$\chi^2 = 5.01$	5	5	0.964	$\chi^2 = 0.28$	
Exploration	3	4	6	6			2		1			3	2			
Competition	2	2	2	2			2		7			6	7			
Socialization	19	14	14	14			15		9			9	9			
QGU-B	26.75 (9.68)	25.05 (8.99)	46.52 (13.40)	46.52 (13.40)	<0.001	F = 29.73	26.44(1.89)	<0.001	47.30(1.63)	<0.001	t = 8.38	45.96(1.49)	46.96(1.81)	0.680	t = 0.42	
IGDS9-SF	17.82 (4.18)	18.64 (4.79)	35.44 (3.07)	35.44 (3.07)	<0.001	F = 151.81	17.30(0.84)	<0.001	28.48(0.75)	<0.001	t = 9.90	31.52(1.31)	29.13(1.18)	0.152	t = 1.76	
VAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	6.09(0.28)	6.35(0.27)	0.507	t = 0.67	
DSM-5	1.86 (1.27)	2.64 (2.01)	7.16 (1.49)	7.16 (1.49)	<0.001	F = 82.64	1.70(0.24)	<0.001	6.78(0.20)	<0.001	t = 16.38	7.30(0.26)	7.13(0.26)	0.636	t = 0.37	
PHQ-9	6.50 (3.63)	7.95 (3.76)	10.72 (6.04)	10.72 (6.04)	0.005	F = 5.64	6.26(0.74)	0.014	9.83(1.18)	0.014	t = 2.57	10.52(0.92)	10.70(1.44)	0.919	t = 0.10	
GAD-7	4.64 (3.71)	5.77 (3.26)	9.00 (4.84)	9.00 (4.84)	<0.001	F = 8.21	4.52(0.81)	0.017	7.83(1.05)	0.017	t = 2.49	8.70(0.63)	7.96(1.20)	0.588	t = 0.55	
Post-intervention sensation of current intensity	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	2.07(0.21)	2.07(0.23)	>0.999	t < 0.001	

IGD: Internet gaming disorder; RGU: recreational game users; HC: healthy controls; QGU-B: Questionnaire of Gaming Urges-Brief; IGDS9-SF: 9-item Internet gaming disorder scale-short form; DSM-5: Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition; PHQ-9: Patient Health Questionnaire-9; GAD-7: Generalized Anxiety Disorder-7; NA: not applicable.

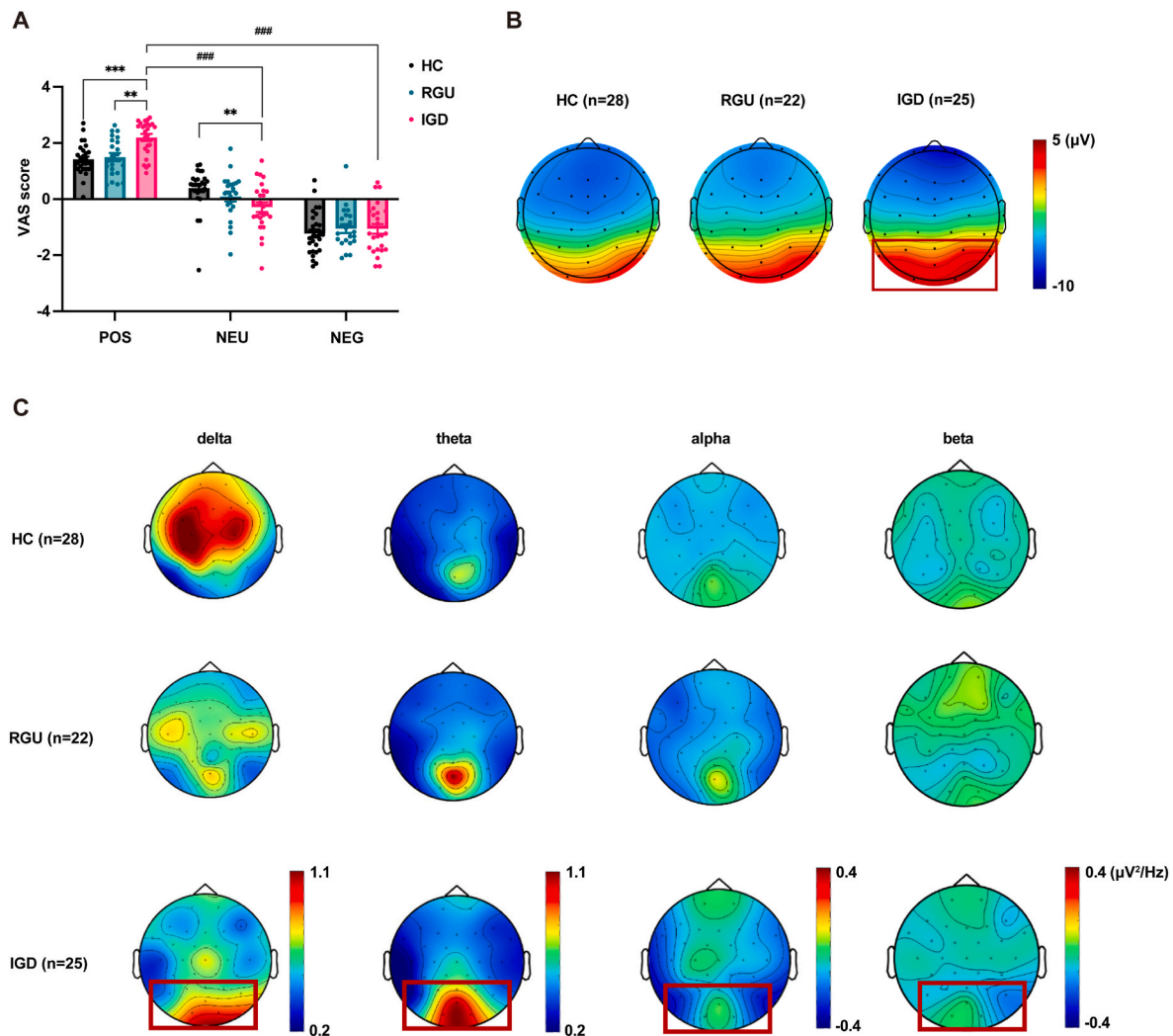


Fig. 2. The differences of the P300 components under game cues among three groups. (A) The VAS craving score under different cues among 3 groups in game cue reactivity task. (B) The topographic map of P300 component (300–500 ms) of parieto-occipital lobe under positive game cue among 3 groups. Red represents high amplitude, and blue represents low amplitude. (C) The topographic map of P300 component (300–500 ms) band power under positive game cue among 3 groups. Red represents high energy and blue represents low energy. The red boxes showed the electrode sites with P300 components in IGD group, which includes a total of 8 electrode sites in the parieto-occipital lobe portion (P3, P4, P7, P8, O1, O2, Pz, POz). IGD: Internet gaming disorder; RGU: recreational game user; HC: healthy control. POS: positive gaming cues; NEG: negative gaming cues; NEU: neutral cues. *: Comparison of VAS craving ratings in the IGD group vs. the other two groups under each condition; #: Comparison of craving ratings for positive game cues vs. the other two cue types within the IGD group; **: $p < 0.01$; ***: $p < 0.001$; ###: $p < 0.001$.

Validation in D2 confirmed the robustness of the D1 P300-based model (accuracy: 80.9 %, sensitivity: 82.2 %, specificity: 78.3 %, AUC = 0.864) for distinguishing IGD from HC (Fig. 3C). Internal validation of the cue-induced P300-based model within D2 yielded comparable performance (accuracy: 78.3 %, AUC = 0.801), demonstrating reproducibility (Fig. 3D).

Weighted feature analysis identified the Pz electrode as the top contributor of the SVM model in both datasets (Fig. 3E–G). Removing Pz features in D1 reduced IGD–HC classification accuracy by 11.3 %, exceeding the impact of other electrodes (3.7–9.4 %), and using Pz alone achieved 76.5 % classification accuracy, outperforming other single-electrode models (accuracy: 49.6–69.8 %) (Fig. 3F). IGD had larger P300 component at Pz under positive cues (D1: $F(2, 72) = 4.67$, $p = 0.012$; D2: $t(44) = 6.93$, $p < 0.001$) (Fig. 3H and I).

Spearman correlation analyses revealed strong positive associations between of Pz delta (D1: $r = 0.52$, $p = 0.009$; D2: $r = 0.53$, $p = 0.009$), theta (D1: $r = 0.49$, $p = 0.012$; D2: $r = 0.54$, $p = 0.007$), and alpha (D1: $r = 0.42$, $p = 0.037$; D2: $r = 0.52$, $p = 0.011$) power with QGU-B craving scores in IGD. No such correlations were observed in RGU or HC ($p >$

0.05) (Fig. 3J and K).

3.4. Participant characteristics of the intervention study

Fifty individuals with IGD were randomized into active ($n = 25$) and sham ($n = 25$) intervention groups, with twenty-three participants per group completing all interventions and follow-ups (total $n = 46$) (Fig. 1C). Baseline demographic and gaming-related measures—including age, sex, daily gaming time, QGU-B and VAS craving scores, IGDS9-SF, DSM-5 criteria, PHQ-9, and GAD-7 scores—showed no significant differences between active and sham groups ($p > 0.05$). Blinding efficacy was confirmed, as subjective current sensation ratings did not differ between two groups ($p > 0.05$) (Table 1). Both groups exhibited significant reductions in positive (time main effect: $F(2, 88) = 9.98$, $p < 0.001$) and negative (time main effect: $F(2, 88) = 29.47$, $p < 0.001$) mood over time (Supplementary Fig. 3). However, no significant group main effects or group $\times$ time interactions were observed for anxiety or depression scores (Supplementary Fig. 4), indicating that tDCS did not specifically modulate emotional states.

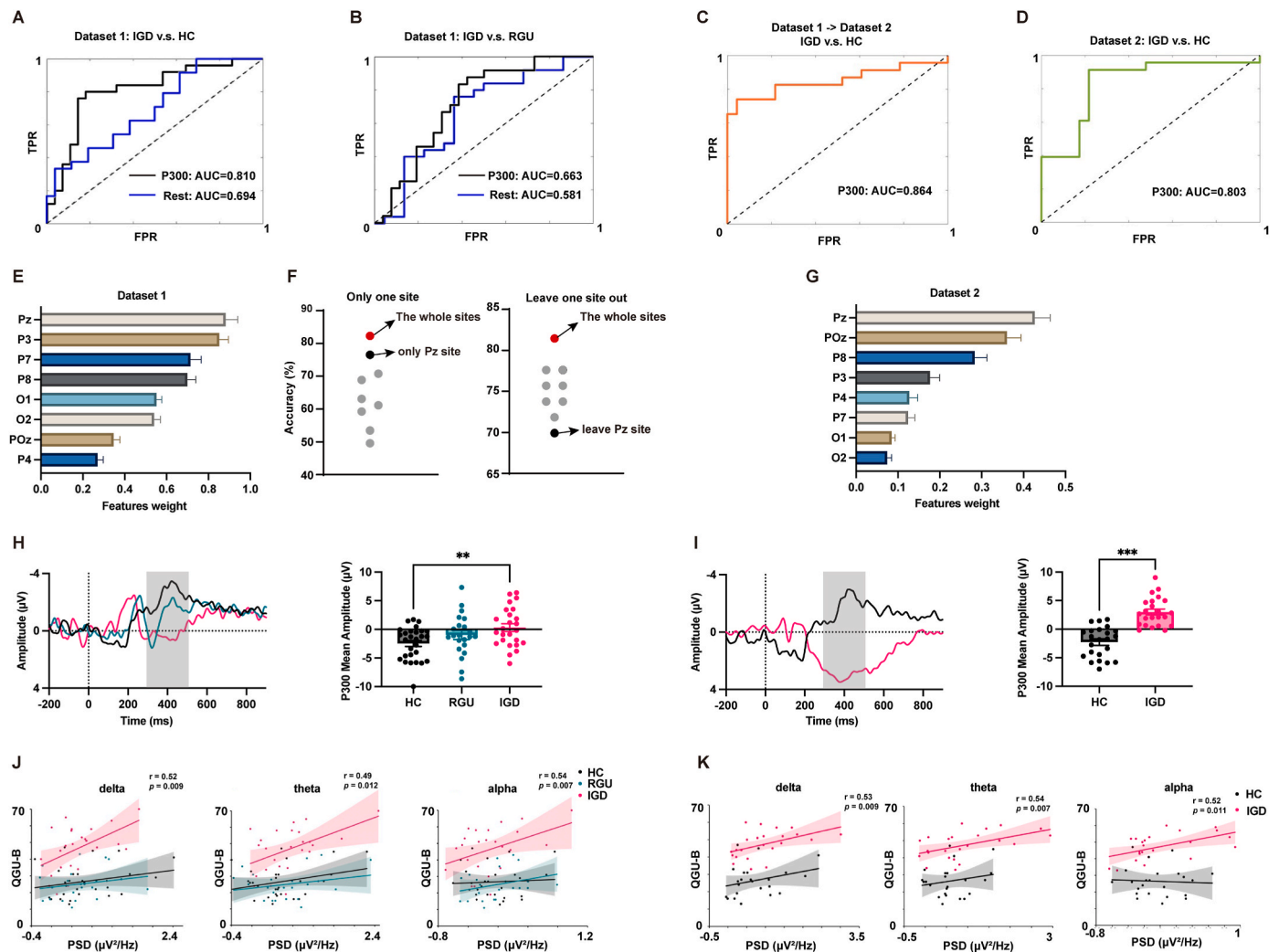


Fig. 3. EEG-based machine learning model to identify IGD. EEG-based machine learning model to identify IGD and HC (A) or RGU (B) in Dataset 1. The blue line is the ROC curve of the model based on the energy of each frequency band at 31 electrode sites throughout the brain in the resting eye-closed state, the black line is the ROC curve of the machine learning model based on the energy of each frequency band of the 8 electrode points under the time window of the P300 of the parieto-occipital lobe under positive game cues. C, D: EEG-based machine learning model to identify IGD and HC in Dataset 2. C: Training on Dataset 1 and tested on Dataset 2. E: Weights of each electrode point in the machine learning model for identify IGD and HC based on the energy of each frequency band under positive game cue in Dataset 1 (E) and Dataset 2 (G). F: Accuracy by machine learning for identifying IGD and HC based on the energy of each frequency band under positive game cue with each electrode site separately and excluded in Dataset 1. The red points represented all the electrode sites, the black points represented Pz site, and the gray points represented other 7 electrode points. The P300 at Pz electrode site under positive game cue among three groups in Dataset 1 (H) and Dataset 2 (I). The horizontal axis is time (ms), the vertical axis is amplitude (μV), and the time point 0 ms is the time point at which the cues appeared. The shaded area is the time window (300–500 ms) in which the P300 component appeared. The associations between power in each frequency band and QGU-B craving scores at Pz in the P300 time window under positive game cue for three groups in Dataset 1 (J) and Dataset 2 (K). IGD: Internet gaming disorder; RGU: recreational game user; HC: healthy control; PSD: power spectral density; QGU-B: questionnaire of gaming urges-brief. ** $p < 0.01$; *** $p < 0.001$.

3.5. Effects of intervention on the craving and gaming time of IGD

To assess the integrity of blinding, we compared the average sensation of current intensity scores between the active and sham groups. An independent-samples *t*-test revealed no significant difference in perceived intensity between the active and sham conditions ($t(44) = 0.92$, $p = 0.362$). This confirms that participants were unable to discern their group assignment, thus validating the success of the blinding procedure.

After controlling for the effects of sex and age, active tDCS significantly reduced craving compared with the sham group (QGU-B: $F(6, 264) = 13.26$, $p < 0.001$; VAS: $F(6, 264) = 8.52$, $p < 0.001$). Post-hoc analyses revealed lower craving in the active group at post_1 week (QGU-B: $p = 0.043$), post_2 week (QGU-B: $p = 0.007$; VAS: $p = 0.046$), and post_3 week (QGU-B: $p = 0.010$; VAS: $p = 0.041$) (Fig. 4A and B). We

also conducted an exploratory subgroup analysis by sex with limited statistical power. Notwithstanding the lack of significance in some comparisons, the observed trends were aligned with the primary analysis, indicating craving reductions in the active groups for males and females (in the Supplemental Results). Daily gaming time decreased significantly in the active group versus sham ($F(4, 176) = 6.58$, $p < 0.001$), with significant post-hoc reductions at post_3 week ($p = 0.011$) and post_4 week ($p = 0.002$) (Fig. 4C).

Then we assessed whether the effects of active tDCS could be generalized to the craving under presented and unpresented gaming cues. IGD subjects reported higher VAS craving for gaming-related cues versus neutral cues ($F(2, 132) = 139.60$, $p < 0.001$) with no difference between presented and unpresented gaming cues at pre-intervention (Supplementary Fig. 5). Post-intervention, active tDCS decreased craving for both presented ($F(3, 132) = 5.08$, $p = 0.002$) and

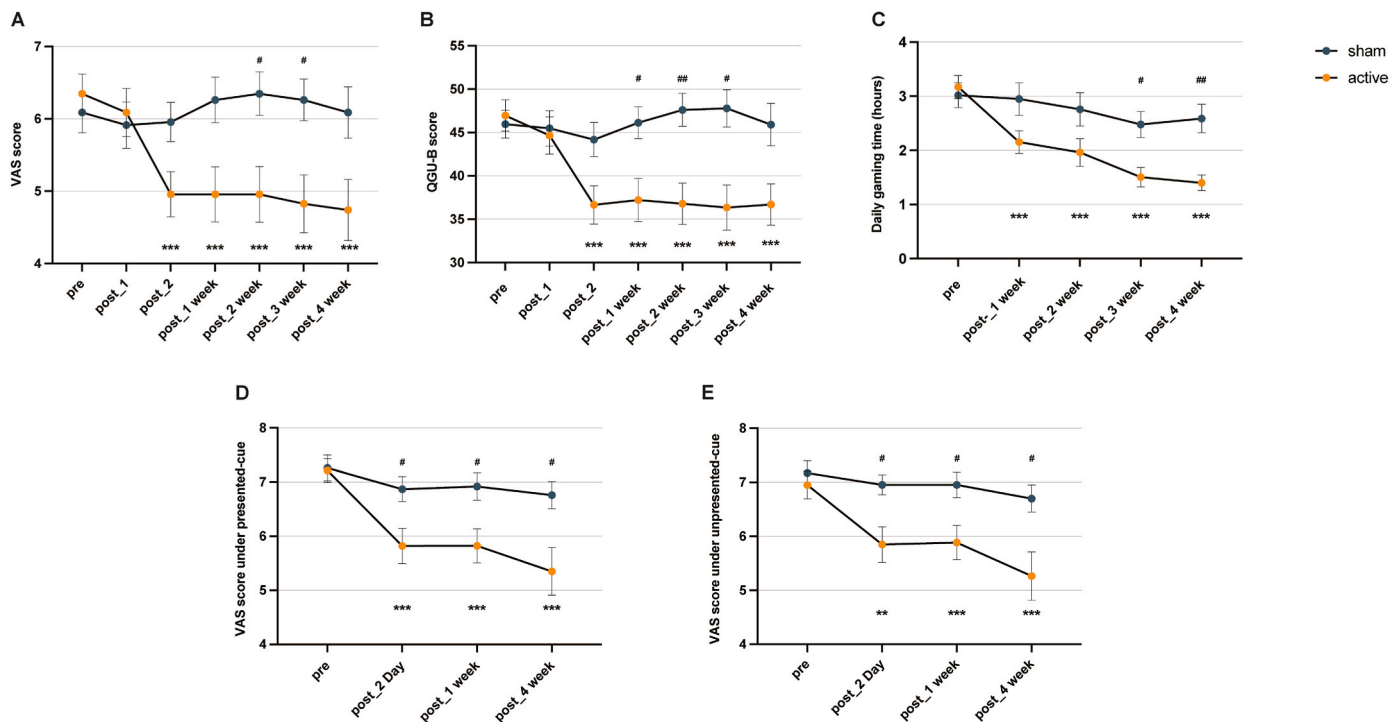


Fig. 4. The effects of intervention on cravings and playtime in IGD. **A:** The changes of VAS score of craving. **B:** The changes of QGU-B score. **C:** The changes of daily gaming time. The changes of VAS score of craving under presented (**D**) and unpresented (**E**) game cue. Active: active tDCS stimulation group; sham: sham tDCS stimulation group; QGU-B: questionnaire of gaming urges-brief; unpresented-cue: the game cues not presented during tDCS; presented-cue: the game cues presented during tDCS. *: active group at each time point compared to pre-intervention in post-hoc analysis; #: comparison between the two groups at each time point in post-hoc analysis; ** $p < 0.01$; *** $p < 0.001$; # $p < 0.05$; ## $p < 0.01$.

unpresented ($F(3, 132) = 3.86, p = 0.011$) gaming cues at post_2 Day ($p = 0.049$), post_1 week ($p = 0.039$), and post_4 week ($p = 0.034$) compared to sham (Fig. 4D, E and Supplementary Fig. 6).

3.6. Effects of intervention on cue-evoked P300 of IGD

No significant group $\times$ time interactions were observed for P300 amplitudes at Pz during exposure to presented gaming cues (Fig. 5A). However, active tDCS significantly reduced P300 amplitudes for unpresented gaming cues (group $\times$ time interaction: $F(3, 132) = 2.87, p = 0.039$), with post-hoc decreases at post_2 Day ($p = 0.047$), post_1 week ($p < 0.001$), and post_4 week ($p = 0.046$) compared to baseline. At post_1 week, P300 amplitudes of the active group were significantly lower than sham ($p = 0.022$) (Fig. 5B). Spearman correlation analysis revealed a significant positive association between P300 amplitude reduction and craving reduction in the active group ($r = 0.56, p = 0.006$), but not in the sham group (Fig. 5C).

Time-frequency analysis demonstrated that alpha power at Pz for presented cues was significantly decreased in the active group (group $\times$ time interaction: $F(3, 132) = 2.71, p = 0.048$), especially at post_1 week ($p = 0.011$) (Fig. 5D–F). For unpresented cues, active tDCS reduced delta (group $\times$ time interaction: $F(3, 132) = 2.75, p = 0.045$) and theta (group $\times$ time interaction: $F(3, 132) = 2.71, p = 0.048$) power at post_1 week (delta: $p = 0.006$; theta: $p = 0.029$) and post_4 week (delta: $p = 0.049$) (Fig. 5G–I and Supplementary Fig. 7).

Notably, baseline delta power at Pz during gaming cues emerged as the sole significant predictor of craving reduction at post_4 week follow-up among P300 components, showing inverse correlations with both VAS ($r = -0.58, p = 0.004$) and QGU-B scores ($r = -0.60, p = 0.003$) (Supplementary Fig. 8). This suggests individuals with lower pre-intervention delta power achieved greater craving reduction.

3.7. Effects of intervention on cognitive performance of IGD

No differences in cognitive performance (SST, MID, IGT) were observed between the active and sham tDCS groups at pre-intervention (Supplementary Table 1). In SST, no significant group $\times$ time interactions were found for Go trial accuracy, Go reaction time, Stop trial accuracy, or Stop trial reaction time (Supplementary Fig. 9). In MID task, also no significant group $\times$ time interactions were found for sensitivity (d') and reaction time in all conditions (Supplementary Fig. 10). Interestingly, in IGT, significant group $\times$ time interactions emerged for net score at post_1 week ($F(5, 220) = 2.39, p = 0.039$) and post_4 week ($F(5, 220) = 2.85, p = 0.016$). Post-hoc tests indicated that the active group exhibited a higher net score than the sham at post_1 week (Block 1: $p = 0.013$) and post_4 week (Block 6: $p = 0.049$), demonstrating progressive decision-making enhancement (Supplementary Fig. 11).

4. Discussion

This study employed machine learning to identify EEG biomarkers for IGD and evaluated the effectiveness of targeted tDCS interventions based on these neural signatures. We established the parieto-occipital P300 activation during positive game-cue exposure as a robust electrophysiological marker for both IGD diagnosis and craving. Our randomized, double-blind, sham-controlled trial revealed that Pz-targeted cathodal tDCS with simultaneously cue exposure significantly reduced craving, gaming time, cue-induced parietal hyperactivity and decision-making abnormality in IGD patient. The treatment effects persisting at one-month follow-up and was generalized to unpresented cue. Also, we identified cue-evoked delta power at Pz as a significant predictor for treatment effect. These results provide a foundation for developing precision neuromodulation therapies for behavioral addictions.

Our machine learning models revealed that resting-state EEG features alone achieved only moderate to low classification accuracy (~60

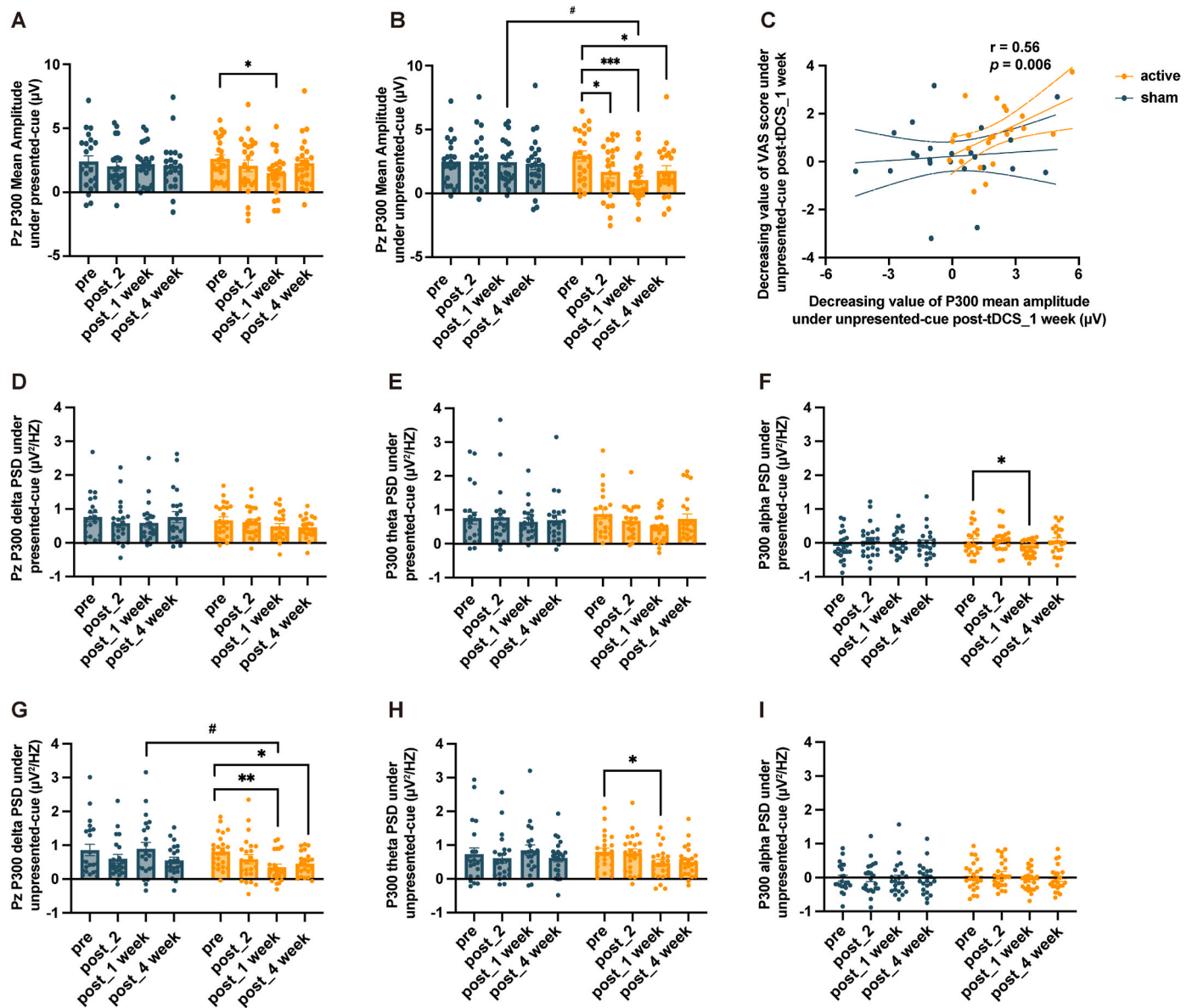


Fig. 5. The effects of intervention on cue-evoked P300 in IGD. The change of P300 mean amplitude under presented (A) and unrepresented (B) game cues. C: The association between decreased P300 amplitude at Pz and decreased VAS craving score under unrepresented game cues one week after tDCS. The change of P300 band power under presented (D ~ F) and unrepresented (G ~ I) game cues. Active: active tDCS stimulation group; sham: sham tDCS stimulation group; unrepresented-cue: the game cues not presented during tDCS; presented-cue: the game cues presented during tDCS. *: active group at each time point compared to pre-intervention in post-hoc analysis; #: comparison between the two groups at each time point in post-hoc analysis; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; # $p < 0.05$.

%) for IGD diagnosis, that falls substantially below the ~80 % accuracy achieved with cue-evoked EEG features and has limited clinical utility. This performance differential highlights the importance of state-dependent, task-based biomarkers for identifying clinically relevant neural targets in neuromodulation interventions. The superior performance of cue-evoked features underscores the critical importance of engaging disease-relevant neural processes during assessment to identify meaningful intervention targets. Research on large sample databases has shown that task state fMRI is superior to resting state fMRI in capturing behavior related brain function information, especially when the task is related to behavioral indicators in cognitive constructs [40].

The visual-evoked P300 component, generated from Pz activity, reflects attentional allocation and working memory processes [41]. Addiction researches consistently demonstrate enhanced P300 amplitudes under drug cues (e.g., cannabis, nicotine, and opioids), indicating attentional bias toward addiction-related stimuli [42–44]. Similarly,

fMRI studies reveal increased parieto-occipital activation during gaming cues in problematic gamers [25]. Our findings demonstrate that enhanced parietal P300 responses to gaming cue exposure serve as a reliable neurophysiological biomarker of IGD, marking dysregulated attentional processing of gaming-related stimuli.

P300 event-related oscillations (EROs) primarily comprise delta and theta bands. Studies have reported altered theta/delta activity in alcohol dependence during the P300 window [45,46]. Alpha-band desynchronization within this window reflects cognitive resource engagement, as observed in food cues responses among overweight individuals [47,48]. Our machine learning models confirmed that Pz delta, theta, and alpha oscillations as craving-correlated biomarkers across datasets. These findings align with prior reports linking P300 amplitudes to opioid craving and alpha oscillations to food craving [42,48]. These findings parallel nicotine addiction studies where P300 normalization predicted craving reduction, positioning parietal EROs as both diagnostic markers

and therapeutic targets [26]. Crucially, baseline P300 delta-band power predicted treatment response, highlighting its potential as a prognostic biomarker for personalized IGD neuromodulation.

Our study advances the field of neuromodulation for IGD by moving beyond the predominant dorsolateral prefrontal cortex (DLPFC)-targeted approach toward biomarker-guided neuromodulation [49]. While anodal DLPFC stimulation has shown promise in reducing craving [11, 21–24,49], our findings demonstrate that parietal-targeted cathodal tDCS represents a paradigm shift from conventional prefrontal excitation toward targeted normalization of pathological network activity. This approach is grounded in emerging evidence of parietal dysfunction in IGD and the well-established inhibitory effects of cathodal stimulation on cortical excitability [20,25].

The use of cathodal tDCS at Pz was motivated by EEG evidence of parietal hyperactivity in IGD and the well-documented inhibitory effects of cathodal stimulation on cortical excitability [50]. This approach aligns with studies showing that cathodal tDCS reduces craving when applied to hyperactive nodes in addiction networks [51]. Our findings further refined this rationale. We identified a pathological multi-band hyperactivity (delta, theta, alpha) in IGD, which collectively represented the intervention target. While transcranial alternative electric stimulation (tACS) excels at frequency-specific modulation, it is less suited for concurrently normalizing multiple distinct bands. In contrast, the broadband inhibition induced by cathodal tDCS is inherently suited to dampen such multi-spectral pathology, justifying its selection and explaining its efficacy.

The robust and sustained therapeutic effects observed in this study likely stem from the synergistic integration of two fundamental mechanisms: the strategic targeting of a key network hub and the precise timing of neuromodulation. The parietal cortex represents an ideal neuromodulation target due to its dual role as a critical hub within both the dorsal attention network (DAN) and fronto-parietal control network (FPN) [52,53]. By inhibiting the hyperactive parietal node, our intervention may have rebalanced the FPN, indirectly ‘releasing’ the DLPFC from aberrant bottom-up drive and thereby restoring its capacity for top-down inhibitory control. This network-level mechanism is strongly supported by our observed improvements in decision-making on the IGT, a function highly dependent on prefrontal integrity [54]. This strategic anatomical position enables it to modulate the critical interplay between bottom-up salience processing and top-down regulatory processes—a balance that is fundamentally disrupted in IGD [55–57]. Parietal modulation producing network-wide effects across fronto-parietal circuits governing attentional control and craving regulation [58].

Beyond target selection alone, the precise synchronization of cathodal stimulation with cue exposure constitutes a second crucial mechanism underlying our intervention’s efficacy. This temporal precision leverages the well-established principle of activity-dependent plasticity [59], whereby applying inhibitory stimulation during active engagement of cue-reactivity circuits enables selective weakening of pathological cue-reward associations at the synaptic level. This approach is supported by substantial evidence that cue-synchronously tDCS produces more robust and lasting neural modulation compared to tDCS alone [60]. The combination of strategic target engagement and precise temporal application creates a powerful therapeutic approach. This dual mechanism explains how brief, precisely-timed intervention can yield durable benefits in addiction [26,61], offering an efficient therapeutic approach that minimizes intervention burden while maximizing lasting benefits through fundamental reconfiguration of maladaptive neural circuits.

Notably, our study addressed a key limitation in prior addiction research by including around 40 % female participants, significantly improving gender balance over prior IGD tDCS trials [24,62]. The therapeutic effects of Pz-targeted tDCS on craving and gaming time remained highly significant after controlling for sex as a covariate, suggesting a broad applicability of this intervention. Exploratory analyses by sex showed promising trends but were limited by sample size;

thus, the question of superior efficacy in a particular sex remains to be conclusively answered in future dedicated studies.

Moreover, this study has several limitations. First, the generalizability of the machine learning findings may be constrained by the sample size and the cohort’s specificity to adult university students primarily playing Honor of Kings. Second, while we recorded gaming time, the lack of data on total screen time and alternative activities precludes a full understanding of the behavioral context mediating the long-term outcomes. Third, future studies should incorporate more granular metrics, such as exact in-game rank, which could help clarify the relationship with pathological EEG signatures while controlling for other confounders. Fourth, while the classification model achieved good accuracy, the ~19 % of misclassified cases represent a valuable source of neurobiological heterogeneity that our current sample size is underpowered to investigate robustly. Therefore, future studies should incorporate granular metrics like in-game rank, combine tDCS with simultaneous fMRI to visualize network modulation, validate our EEG biomarkers in larger cohorts to enable personalized protocols, and directly compare parietal versus DLPFC stimulation to define optimal targets.

In summary, this study identified the cue-evoked parieto-occipital P300 component as a robust electrophysiological marker for IGD diagnosis and craving assessment. Our randomized control trial demonstrated that Pz-targeted cathodal tDCS synchronized with gaming cue exposure produced significant, sustained (≥ 1 month) reductions in gaming time, craving and pathological neural activation of IGD. These findings establish cue-synchronized tDCS as an effective intervention approach and position the Pz as a novel therapeutic target for IGD neuromodulation.

CRedit authorship contribution statement

Yixuan Song: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuchen Huang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Qihong Zheng:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Xiaoqin Yang:** Writing – review & editing, Investigation, Data curation. **Yang Guo:** Writing – review & editing, Investigation. **Yanan Geng:** Methodology, Software. **Huixing Gou:** Writing – review & editing, Investigation, Data curation. **Junjie Bu:** Writing – review & editing, Validation. **Tianye Jia:** Writing – review & editing, Validation. **Guangdong Zhou:** Writing – review & editing, Validation, Resources. **Lin Lu:** Writing – review & editing, Supervision, Resources. **Jie Shi:** Writing – review & editing, Supervision, Resources. **Yan Sun:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.brs.2025.2016>.

org/10.1016/j.brs.2025.10.025.

Data availability

Data can be requested from the corresponding author. Data use and requests underlie the publication policy of the multicentre GET. FEED-BACK.GP-RCT.

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